

ASIR INTESAR TUSHAR

209 Ayres hall, Knoxville, USA. ✉ aintesar@vols.utk.edu, ☎ +1 (865) 443-9282

EDUCATION

Ph.D. in Mathematics <i>University of Tennessee - Knoxville</i>	Knoxville, USA (expected) August 2027
· Advisor: Dr. Ioannis Sgouralis	
M.S. in Mathematics, minor in Computational Science <i>University of Tennessee - Knoxville</i>	Knoxville, USA Spring 2026
M.S. in Mathematics <i>University of Dhaka</i>	Dhaka, Bangladesh July 2020
B.S. in Mathematics <i>University of Dhaka</i>	Dhaka, Bangladesh Aug 2018

RESEARCH INTERESTS

Bayesian Statistics, Computational Statistics, Markov Chain Monte Carlo Methods, Statistical Learning, Uncertainty Quantification, Bayesian Nonparametrics, Point Cloud Data Analysis, and Hidden Markov Models

PUBLICATIONS

- A. I. Tushar** and I. Sgouralis. *Bayesian methods and Markov chain Monte Carlo algorithms for curve reconstruction and point cloud data analysis*. arXiv:2608.26490, 2026. Pre-print
- J. A. Khan and **A. I. Tushar**. *Numerical Approximations of a Nonlinear Volatility Model with European Options*. *GANIT: Journal of Bangladesh Mathematical Society*, 42(1), 50–68, 2022. doi:10.3329/ganit.v42i1.61000.

RESEARCH EXPERIENCE

- | | |
|--|---|
| Graduate Research Assistant – Sgouralis Research Group
<i>University of Tennessee, Knoxville</i> | Knoxville, TN
January 2026 – Present |
|--|---|
- Developed Bayesian models and tailored Markov chain Monte Carlo algorithms for reconstructing closed curves from noisy point-cloud data, with uncertainty quantification for the recovered geometry.
 - Developing Bayesian nonparametric classification and hidden Markov models for disease detection in dairy cows using sensor-derived animal health data.
- | | |
|--|--|
| Research Fellow
<i>Bose Centre for Advanced Study and Research</i> | Dhaka, Bangladesh
February 2020 – June 2020 |
|--|--|
- Studied numerical approximations of nonlinear Black–Scholes equations under several volatility models using finite-difference and finite-element methods, resulting in a peer-reviewed publication.

TEACHING EXPERIENCE

- | | |
|---|--|
| Graduate Teaching Associate
<i>University of Tennessee, Knoxville</i> | Knoxville, TN
August 2022 – Present |
|---|--|
- **Instructor of record for 16 sections:**
 - **MATH 151: Mathematics for the Life Sciences** — Fall 2026 (2), Fall 2025 (2), Spring 2025 (1), Fall 2024 (2)
 - **MATH 152: Calculus for the Life Sciences** — Spring 2026 (1)

- **MATH 113: Mathematical Reasoning** — Spring 2024 (2), Fall 2023 (2)
- **MATH 123: Finite Mathematics** — Spring 2023 (2), Fall 2022 (2)

Graduate Teaching Assistant
University of Tennessee, Knoxville

Knoxville, TN
August 2021 – July 2022

- Served as a teaching assistant for **MATH 125: Basic Calculus** (Spring 2022) and **MATH 119: College Algebra** (Fall 2021).

Lecturer
Ahsanullah University of Science and Technology

Dhaka, Bangladesh
December 2020 – June 2021

- Instructor of record for five sections: **MATH 1107: Calculus I** (3 sections) and **MATH 0231: Differential Equations** (2 sections).

Trainer
Bangladesh Mathematical Olympiad Committee

Bangladesh
2016 – 2019

- Led training sessions in number theory and combinatorics and supported national-level mathematical olympiad activities.

Mentor
Bangladesh Mathematical Olympiad Committee

Bangladesh
2013 – 2016

- Trained participants, developed problems, graded solutions, and supported divisional mathematical olympiads.

TALKS

Curve Reconstruction from Point Cloud Data
2026 SIAM Annual Meeting (AN26), Cleveland, OH

July 2026

Markov Chain Monte Carlo Algorithms for Curve Reconstruction and Point Cloud Analysis
UTK SIAM Graduate Student Research Showcase, University of Tennessee, Knoxville

March 2026

Efficient Bayesian Methods for Curve Reconstruction and Point Cloud Analysis

51st Annual New York State Regional Graduate Mathematics Conference, Syracuse University March 2026

Numerical Investigation of the Black–Scholes Equation

Department of Mathematics, University of Dhaka, Dhaka, Bangladesh

December 2019

Mercury's Perihelion Precession: A Classical Test of General Relativity

Department of Mathematics, University of Dhaka, Dhaka, Bangladesh

August 2018

CONFERENCES AND WORKSHOPS

International Congress of Mathematicians (ICM 2026)
Philadelphia, PA

July 2026

SIAM Annual Meeting 2026
Cleveland, Ohio

July 2026

AstroAI Workshop 2026
Center for Astrophysics — Harvard & Smithsonian (Virtual)

June 2026

The 54th John H. Barrett Memorial Lectures: Mathematical Foundations for Computational and Data-Driven Scientific Discovery

University of Tennessee, Knoxville

May 2026

105th Annual Meeting of the MAA Southeastern Section <i>University of North Alabama, Florence, AL</i>	March 2026
51st Annual New York State Regional Graduate Mathematics Conference <i>Syracuse University, Syracuse, NY</i>	March 2026
Bayesian Inverse Problems and Uncertainty Quantification <i>ICERM, Brown University, Providence, RI</i>	March 2026
SIAM Southeastern Atlantic Section Meeting <i>University of Tennessee, Knoxville</i>	March 2025
103rd Annual Meeting of the MAA Southeastern Section <i>University of Tennessee, Knoxville</i>	March 2024
International Mathematics Conference <i>University of Dhaka</i>	December 2019

HONORS, AWARDS & FELLOWSHIPS

Research and Travel Awards

- **First Place**, UTK SIAM Graduate Student Research Showcase
University of Tennessee, Knoxville 2026
- AMS–NSF ICM 2026 Travel Grant, International Congress of Mathematicians 2026
- SIAM Student Travel Award, 2026 SIAM Annual Meeting (AN26) 2026
- Graduate Student Senate Travel Award, University of Tennessee, Knoxville
SIAM Conference on the Life Sciences (LS26) 2026
- NSF-Supported Travel Award, 51st Annual New York State Regional Graduate Mathematics Conference
2026
- Participant Travel Support, Bayesian Inverse Problems and Uncertainty Quantification Workshop, ICERM
2026
- Departmental Travel Award, Department of Mathematics, University of Tennessee, Knoxville 2026

Fellowships and Competitive Assistantships

- Graduate School Fellowship, University of Tennessee, Knoxville 2024–2026
- Academic Performance Assistantship, University of Tennessee, Knoxville 2023
- Academic Excellence Fellowship, University of Tennessee, Knoxville 2021–2022
- Bose Research Fellowship, Bose Centre for Advanced Study and Research in Natural Sciences, University
of Dhaka 2020
- National Science and Technology (NST) Fellowship for M.S. Thesis Research, Ministry of Science and
Technology, Bangladesh 2018–2019

Teaching and Academic Honors

- Eaves Teaching Award Finalist, University of Tennessee, Knoxville 2023
- Trust Bank Merit Award 2014–2018
- Honors Merit Award, University of Dhaka 2017
- Academic Merit Award, Shahidullah Hall, University of Dhaka 2014–2017

Competition Honors

- Second Place (First Runner-Up), Inter-University Astrophysics and Astronomy Olympiad 2016
- Champion, Intra-Departmental Mathematics Olympiad 2014

TECHNICAL SKILLS

Programming and Scientific Computing: MATLAB, Python, Fortran, Mathematica

Document Preparation: $\text{\LaTeX}$

Languages: Bengali (native), English (fluent)

PROFESSIONAL SERVICE AND LEADERSHIP

Graduate Student Senator, Department of Mathematics

University of Tennessee, Knoxville

2026–Present

Co-organizer, UTK Student-led Graduate Seminar (MATH 597)

University of Tennessee, Knoxville

2025–Present

Co-organizer, Minisymposium on *Application-Driven Computational Modeling: Stochastic, Probabilistic, and Data-Driven Methods*

2026 SIAM Annual Meeting (AN26), Cleveland, OH

2026

Reviewer, *Journal of Computational and Theoretical Transport*

2026

Judge, 30th Annual Exhibition of Undergraduate Research and Creative Achievement (EURēCA)

University of Tennessee, Knoxville

2026

PROFESSIONAL MEMBERSHIPS

American Mathematical Society (AMS)

Society for Industrial and Applied Mathematics (SIAM)

Bangladesh Student Association (BSA)

Mathematical Association of America (MAA)

Bangladesh Mathematical Society (BMS)

UTK Table Tennis Club